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Soil water balance and nitrate leaching in winter wheat–summer maize double-cropping systems with different irrigation and N fertilization in the North China Plain

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Summary—Zusammenfassung

Field experiments over a 3 y period were conducted in a winter wheat-maize double-cropping system at the Dongbeiwang Experimental Station, Beijing, China. Three different treatments of irrigation (sprinkler "suboptimal" and "optimized"; conventional flood irrigation) and N fertilization (none, according to N_{min} soil tests, conventional) were studied with respect to effects on soil water balance, nitrate leaching, and grain yield. Under sprinkler irrigation, evaporation losses were higher due to a more frequent water application. On the other hand, in this treatment nitrate leaching was smaller as compared to flood irrigation, where abundant seepage fluxes >10 mm d^{-1} along preferential flow paths occurred. For quantifying nitrate leaching, passive samplers filled with ion-exchange resins appeared to be better suited than a method which combined measurements of suction-cup concentrations with model-based soil water fluxes. As a result of the more balanced percolation regime (compared to that under conventional flood irrigation), there was a tendency of higher salt load of the soil solution in the rooting zone. Given a seepage rate of 50 mm, a winter wheat grain production of 5–6 t ha^{-1} required a total water addition of about 430 mm. Fertilizer treatments >100 kg N ha^{-1} did not result in any additional yield increase. An even balance between withdrawing and recharge of groundwater cannot be achieved with "optimized" irrigation, but with a reduction of evapotranspiration losses, adapted cropping systems, and/or by tapping water resources from reservoirs in more distant areas with surpluses.

Key words: Calcaric Cambisol / soil water balance / nitrate leaching / grain production / North China Plain

Bodenwasserhaushalt und Nitratauswaschung in Winterweizen-Sommermais-Anbausystemen mit unterschiedlicher Bewässerung und N-Düngung in der nordchinesischen Tiefebene

Feldexperimente mit einer Winterweizen-Sommermais-Doppelfruchtfolge erfolgten über einen dreijährigen Zeitraum auf dem Versuchsfeld Dongbeiwang bei Peking, China. Untersucht wurde die Auswirkung von je drei Varianten der Bewässerung (Sprinkler „suboptimal" und „optimiert" sowie konventionelle Überstaubewässerung) und der N-Düngung (ohne, basierend auf N_{min}-Bodenuntersuchungen, konventionell) auf Bodenwasserhaushalt, Nitratauswaschung und Kornertrag. Bei Sprinklerbewässerung entstanden durch die häufigere Wasserzufuhr größere Evaporationsverluste. Andererseits war hier die Nitratauswaschung deutlich geringer als bei Überstaubewässerung, bei der häufig Sickerwasserströme von >10 mm d^{-1} entlang präferenzieller Fließwege auftraten. Zur Erfassung der N-Auswaschung erwiesen sich Ionenaustauscher-Passivsammler als besser geeignet als die Kombination von Saugkerzenlösungen und modellberechneten Sickerraten. Unter dem bei Beregnung im Vergleich zur Überstaubewässerung ausgeglicheneren Sickerungsregime ergaben sich Hinweise auf eine Salzanreicherung im Wurzelraum. Die Erzeugung von 5–6 t ha^{-1} Winterweizenkornertrag erfordert bei 50 mm Sickerung eine Gesamtwasserzufuhr von ca. 430 mm. Düngergaben >100 kg N ha^{-1} führten zu keiner weiteren Ertragssteigerung. Eine ausgeglichene Bilanz zwischen Neubildung und Entnahme von Grundwasser ist nicht mit „optimierter" Bewässerung, sondern nur über eine Reduzierung der Verdunstungsverluste, angepasste Fruchtfolgen und/oder Wasserzufuhr aus weiter entfernten Gebieten mit Wasserüberschuss zu erreichen.

1 Introduction

The North China Plain is a traditional high-yielding crop-production area. However, the highly variable summer-monsoon precipitation has a strong effect on the productivity. To keep-up with population growth, raising consumption, and losses of arable land around the economically booming conurbations, the traditional rain-fed farming has been replaced by irrigated cropping systems in order to increase crop production. Combined with mechanization and mineral fertilization, the productivity has increased manifold in recent decades (*Tso* et al., 1998). A serious side-effect of that development consists in the fact that groundwater levels in the North China Plain have been dropping continuously (*Kendy* et al., 2003). Furthermore, an excessive use of N fertilizers has led to elevated nitrate concentrations in the groundwater, which often exceed drinking-water standards (*Zhang* et al., 1996). As a consequence, there is an increasing need for adapted agricultural practices, which ensure a high level of crop production. Simultaneously, they are more efficient with respect to the consumption of irrigation water and fertilizers. In recent years, extensive agricultural research was carried out in the North China Plain, *e.g.*, by *Pereira* et al. (1998, 2003) and *McVicar* et al. (2002).

Within the framework of the Sino-German research co-operation project "Environmental acceptable and sustainable agri-

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culture with high yield and high productivity in the North China Plain (1999–2002)", numerous field experiments were conducted aiming at

- assessing the soil water balance by the use of a computer model;
- quantifying nitrate leaching below the rooting zone as part of the N balance under various N-fertilization and irrigation regimes including the comparison of methods;
- establishing an instrument for irrigation management within agricultural field experiments by implementing measuring devices and recording soil physical properties;
- deriving recommendations for an environment-friendly land use in the study area.

2 Materials and methods

We undertook field experiments during the period July 1999 to June 2002 in winter wheat-summer maize double-cropping systems, which currently are prevailing in the practical agriculture of the North China Plain. The experiment was conducted at the experimental farm of the China Agricultural University (CAU), located at Dongbeiwang in Beijing, China (39°55′ N, 116°20′ E). The site is located in a typical semiarid continental monsoon climate. More than 70% of the average annual rainfall ranging from 500 mm to 650 mm normally fall between June and August. However, the year-to-year variation is large. Mean annual temperature is 11.5°C, the calculated annual potential evaporation is close to 800 mm. Three treatments of irrigation (only in the winter wheat period) and three treatments of N fertilization were conducted on an experimental site of 5 ha (Tab. 1; more details in *Chen*, 2003; *Böning-Zilkens*, 2004; *Mack*, 2005). In the first year, suboptimal irrigation (A) of winter wheat was 184 mm in four sprinkler applications, optimized irrigation (B) was 317 mm in eight sprinkler applications, conventional irrigation (C) was 330 mm in one sprinkler and three flood applications. All irrigation water was derived from a groundwater well on the experimental field. The three N-fertilization treatments of winter wheat in the first year were none (a), 10 to 92 kg ha^{-1} as urea, according to the plot's soil N_{min} content (b), 300 kg ha^{-1} as urea and NH_4HCO_3 (c). In the second year, amounts and application frequencies were in a similar range.

The soil is a Calcaric Cambisol with a silty-loamy texture. The soil water characteristic was determined by field observations (such as hood infiltrometer readings) and laboratory meas-

Table 1: Frequencies and amounts of irrigation and N fertilization.
Irrig. = irrigation, N = N fertilization. Kind and number of applications: s = sprinkler irrigation, f = flood irrigation, u = urea N, m = mineral N as NH_4HCO_3.
Explanation of codes:
A: suboptimal sprinkler irrigation,
B: optimized sprinkler irrigation, keeping soil moisture in the rooting zone between 45% and 80% of the available water capacity (AWC),
C: conventional flood irrigation.
a: without any N fertilization in the present cultivation period, followed by (b),
b: optimized N fertilization, based on the results of N_{min} soil analysis (70–200 kg N ha^{-1} y^{-1}),
c: conventional N fertilization (600 kg N ha^{-1} y^{-1}).

Tabelle 1: Häufigkeit und Mengen der Bewässerung und N-Düngung.
Irrig. = Bewässerung, N = N-Düngung. Art und Anzahl der Anwendungen: s = Sprinklerbewässerung, b = Überstaubewässerung, u = Harnstoff-N, m = mineralischer Stickstoff als NH_4HCO_3.
Codes:
A: suboptimale Sprinklerbewässerung,
B: optimierte Sprinklerbewässerung zur Gewährleistung von Bodenwassergehalten zwischen 45 und 80 % der nutzbaren Feldkapazität (AWC) im Wurzelraum,
C: konventionelle Überstaubewässerung.
a: ohne N-Düngung in der aktuellen Anbauperiode, gefolgt von (b),
b: optimierte N-Düngung, bemessen anhand der Ergebnisse von N_{min}-Bestimmungen im Boden (70–200 kg N ha^{-1} a^{-1}),
c: konventionelle N-Düngung (600 kg N ha^{-1} a^{-1}).

	First year				**Second year**			
	Maize 1999		**Wheat 99/00**		**Maize 2000**		**Wheat 00/01**	
Code	Irrig. [mm]	N [kg ha^{-1}]	Irrig. [mm]	N [kg ha^{-1}]	Irrig. [mm]	N [kg ha^{-1}]	Irrig. [mm]	N [kg ha^{-1}]
Aa	10	0	184	0	33	0	230	0
Ab	s	110 mu	ssss	10 u	s	10 u	ssss	62 u
Ac		300 mu		300 mu		300 mu		300 mu
Ba	10	0	317	0	33	0	307	0
Bb	s	110 mu	ssss	80 uu	s	54 u	ssss	70 uu
Bc		300 mu	ssss	300 mu		300 mu	s	300 mu
Ca	10	0	330	0	33	0	345	0
Cb	s	110 mu	sfff	92 uu	s	31 u	sfff	83 uu
Cc		300 mu		300 mu		300 mu		300 mu

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urements with samples from the different soil layers according to *Schlichting* et al. (1995). Additionally, the pedotransfer function "Rosetta" (*Schaap* et al., 1998) was used. A "standard soil profile", including generalized data of five soil profiles within an area of 5 ha and an average field water-retention curve was used to implement the conceptional, one-dimensional water-budget simulation model COUP (*Jansson and Karlberg*, 2001). We used on-site weather data as well as plant-parameter data, which were supplied by the agronomy group of the Sino-German co-operation project (*Böning-Zilkens*, 2004).

For model calibration and irrigation scheduling within the same treatment, soil moisture was measured at depth intervals of 30 cm with nine tensiometers and TDR-probes (type E.S.I./MP-917), respectively, which had been calibrated for the site. TerrAquat® passive samplers (*Bischoff* et al., 1999), filled with ion-exchange resin, were used to quantify the total annual amount of leached nitrate below the rooting zone (ten replicates per plot). In addition, seepage nitrate losses were estimated indirectly by a simple multiplication of the nitrate concentrations analyzed in the soil solution (noncontinuously sampled every two weeks with ceramic suction cups, five replicates) with the calculated soil water fluxes in the corresponding depth of 135 cm. Evaporation, transpiration, interception, and seepage water were calculated using the COUP model.

3 Results and discussion

3.1 Soil water balance and percolation regimes

The entire experimental field at Dongbeiwang is characterized by a distinct high spatial variability of soil parameters, which is typical for floodplain soils (*e.g.*, the percentage of sand ranged between 10% and 70%, saturated water conductivity between 40 to 500 cm d^{-1}). Therefore, the model calibration holds only for an area of approximately 5 ha where we conducted an extensive soil inventory. Correlation coefficients in the range of $r^2 = 0.5$ and average deviations of maximum 3.2 Vol.-% were found between the measured and simulated soil water contents. As repeated inventories have shown, the distribution of irrigation water and soil water contents varied by up to 15% within the same treatment. Accordingly, irrigation management took place with that accuracy. The available water capacity (AWC) in the uppermost 100 cm was determined to be approximately 220 mm.

The "optimized" irrigation treatment B aimed at controlling soil moisture in a range between 45% and 80% of AWC that is considered to be best suitable for winter wheat cultivation. In Fig. 1, the temporal variation of the amount of available water is shown. The goal was reached satisfactorily. Drought stress was avoided during the entire growth period and available soil water in the rooting zone was more balanced because sprinklers were used. This is due to the fact that irrigation amounts per application event were smaller. According to the COUP calculations, the transpiration rates in treatment B were highest, indicating that growth conditions there were the relatively best (Tab. 2). In this "optimized" treatment, a reduction of the irrigation amount of about 10% compared to conventional flood-irrigation treatment C did not have significant effects on grain yield of winter wheat (*Böning-Zilkens*, 2004). Under the given conditions, approximately 430 mm of total water input (irrigation and precipitation, percolation losses of 50 mm included) were required to ensure a winter wheat-grain production of 5–6 t ha^{-1}.

After flood-irrigation events, soil water contents in the rooting zone usually exceeded field capacity. Although total annual water inputs differed by only 10%, the calculated seepage-water amounts were about twice as high as in treatment B with sprinkler irrigation. A considerable amount of irrigation water percolated with an intensity of more than 10 mm d^{-1} (Fig. 2). In order to trace quick percolation, we applied a bromide tracer on the surface before irrigation started. The bromide recovery in 135 cm depth 1 y after application suggested that with the relatively low sprinkler-water-application rate of 6 to 9 mm h^{-1}, seepage along preferential flow paths (mainly earthworm channels) was reduced threefold compared to the flood irrigation treatment C (*Mack*, 2005).

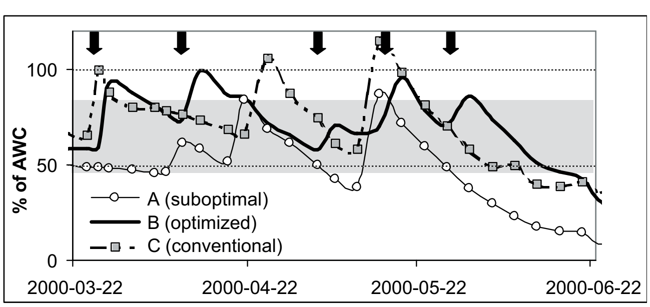

Figure 1: Share of the available water capacity (AWC) in the 0–60 cm soil layer under winter wheat cultivation with the irrigation treatments A, B, C. Arrows indicate irrigation events in treatment B ("optimized"), the preset range of soil moisture to be maintained in this treatment is indicated in grey.

Abbildung 1: Anteile der nutzbaren Feldkapazität (AWC) in der Bodenschicht 0–60 cm bei Winterweizenanbau mit den Bewässerungsvarianten A, B, C. Schwarze Pfeile markieren Bewässerungsereignisse bei Variante B („optimiert"), grau hinterlegt ist der dort vorgegebene Bereich der einzuhaltenden Bodenfeuchte.

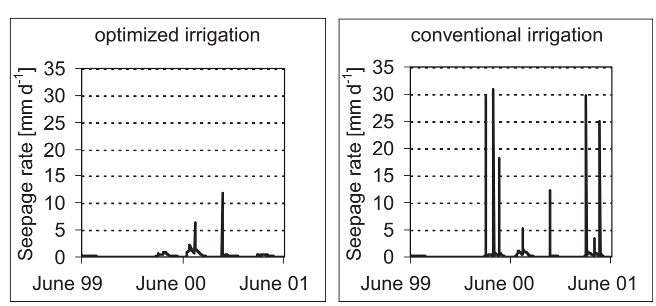

Figure 2: Comparison of percolation regimes under optimized sprinkler irrigation and conventional flood irrigation.

Abbildung 2: Vergleich der Sickerungsdynamik unter optimierter Sprinklerbewässerung und konventioneller Überstaubewässerung.

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Under sprinkler irrigation, soil water below the rooting zone moved downwards at rates of mostly <1 mm d^{-1}. As a result of this more balanced percolation regime (compared to that under conventional flood irrigation), a tendency of higher salt load in the soil solution was observed. Therefore, the leaching rate under "optimized" sprinkler irrigation might have been below the "leaching requirement" (approximately 15% of the total water input with the given salinity of the irrigation water, according to *Hanson* (1993)). During an observation period of 2 y, the electric conductivity of the soil solution at 135 cm depth increased to approximately 1.5 mS cm^{-1} under "optimized" sprinkler irrigation, while it remained constant at 1.0 mS cm^{-1} under the flood irrigation treatment.

In treatments with conventional flood irrigation, the soil surface quickly dried out in periods between the monthly water applications of about 70–100 mm, notably in the dry and windy spring season. Amounts of evaporation water during the winter wheat period increased by approximately 10 mm with the more frequent sprinkler irrigation events. Highest unproductive water losses (up to 3 mm d^{-1}) occurred in the first weeks after maize had been sown (Fig. 3). During that time period, the soil surface was poorly covered with plants, and air temperatures were very high (daily means 25°C–30°C). Simultaneously, frequent rainfalls (summer monsoon) caused a sustained moistening of the upper soil layer. As the water transfer to the atmosphere is increased particularly after winter wheat harvest (mid of June), it would be reasonable to have a straw-mulch cover saving the water reserves in the top soil. This would also be beneficial for the growing maize plants which are sensitive towards drought stress.

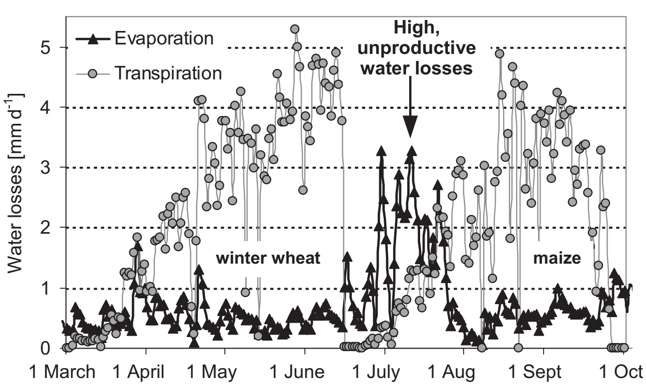

Figure 3: Atmospheric water losses in the winter wheat–summer maize double-cropping system. Simulation with the COUP model for the year 2000.
Abbildung 3: Atmosphärische Wasserverluste in der Winterweizen-Sommermais-Doppelfruchtfolge. Simulation mit dem COUP-Modell für das Jahr 2000.

The individual water-balance parameters under irrigation treatments A, B, and C in the field experiment were compared with three COUP modeled scenarios (Tab. 2). In scenario 0, which represents a bare soil with no plant cover, annual actual evaporation-water losses are expected to be clearly smaller than precipitation as the only water input. Approximately 120 mm of seepage water would have been produced by summer rainfalls, leading to a distinct recharge of the groundwater. With similar demands on soil water supply, maize achieved comparable grain yields as winter wheat in our field experiment, despite of higher evaporation losses and occasional drought stress. Because of its growth in the rainy summer season, maize is much less dependent on irrigation than winter wheat. Scenario M assumes an exclusively rain-fed maize cultivation in summer and a long fallow period from October to June. Although traditional summer-production systems would be best suited to protect groundwater resources, they are not likely to recur in view of the enormous pressure to produce maximum grain yields in present-day densely populated China. In scenario K, flood irrigation with the same total amounts as in C was carried out, adding an excessive amount of 150 mm on April 15th and May 15th, respectively. Such a practice is also used in winter wheat in the

Table 2: Simulated soil water balances in the depth interval 0–135 cm and groundwater balances under different production and irrigation scenarios; average values of three years (1999–2002), all units are [mm], extreme values are bold.
irrigation + precipitation + capillary rise – Δ soil water storage = evaporation + transpiration + interception + seepage.
Δ groundwater = seepage – capillary rise – irrigation (using groundwater)
Explanation of abbreviations:
A: suboptimal sprinkler irrigation,
B: optimized sprinkler irrigation keeping soil moisture in the rooting zone between 45% and 80% of the available water capacity (AWC),
C: conventional flood irrigation,
K: flood irrigation with total amounts as in C but distributed at only two irrigation events,
M: maize cultivation and winter fallow, no irrigation,
0: fallow, no irrigation.
Tabelle 2: Simulierte Bodenwasserbilanzen im Tiefenintervall 0–135 cm sowie Grundwasserbilanzen unter verschiedenen Produktions- und Bewässerungsszenarien; jeweils Dreijahresmittelwerte (1999–2002), alle Angaben in [mm], Extremwerte fettgedruckt.
Bewässerung + Niederschlag + Kapillaraufstieg – Δ Bodenwasservorrat = Evaporation + Transpiration + Interzeption + Sickerung.
Δ Grundwasser = Sickerung – Kapillaraufstieg – Bewässerung (mit Grundwasser)
Abkürzungserklärungen:
A: suboptimale Sprinklerbewässerung,
B: optimierte Sprinklerbewässerung zur Gewährleistung von Bodenwassergehalten zwischen 45 und 80 % der nutzbaren Feldkapazität (AWC) im Wurzelraum,
C: konventionelle Überstaubewässerung,
K: Überstaubewässerung mit Gesamtmengen wie bei C, aber auf nur zwei Ereignisse verteilt,
M: Maisanbau und Winterbrache, keine Bewässerung,
0: Brache, keine Bewässerung.

Treatment	0	M	A	B	C	K
Irrigation	0	0	235	338	**376**	**376**
Precipitation	420	420	420	420	420	420
Capillar rise	0	1	**12**	4	5	9
Δ soil water storage	–14	–14	+33	+23	+21	+27
Evaporation	**286**	178	170	181	171	156
Transpiration	0	132	468	**486**	478	463
Interception	0	18	31	**32**	30	28
Seepage	120	79	31	86	143	**185**
Δ groundwater	+120	+78	–216	**–256**	–238	–200

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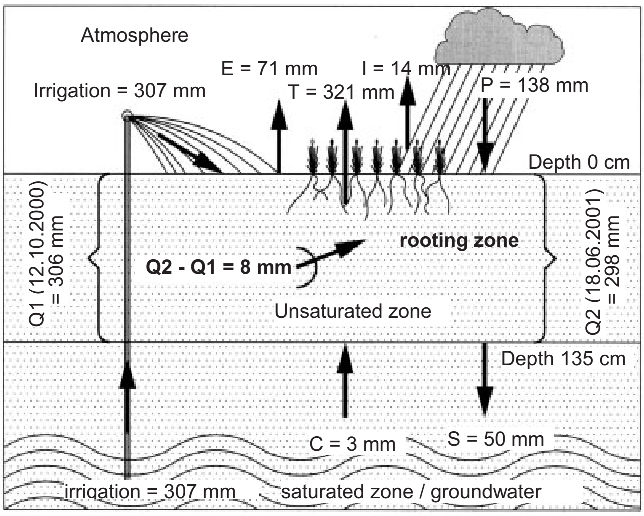

Figure 4: Soil water balance in the unsaturated zone and groundwater balance (field scale) under irrigated winter wheat (treatment B, 2nd year); Irrigation = irrigation and corresponding groundwater pumping, E = evaporation*, I = interception*, C = capillary rise at 135 cm depth*, P = precipitation, Q (...) = soil water storage in the depth interval 0–135 cm at a certain time*, S = seepage at 135 cm depth*, T = transpiration of the wheat stand*; * = modeled using COUP.

Abbildung 4: Bodenwasserbilanz in der ungesättigten Zone und Grundwasserbilanz einer Versuchsparzelle bei bewässertem Winterweizenanbau (Variante B, zweites Beobachtungsjahr); Irrigation = Bewässerung bzw. zugehörige Grundwasserentnahme, E = Evaporation*, I = Interzeption*, C = Kapillaraufstieg in 135 cm Tiefe*, P = Niederschlag, Q (...) = Bodenwasservorrat im Tiefenintervall 0–135 cm zu einem Zeitpunkt*, S = Sickerung in 135 cm Tiefe*, T = Transpiration des Weizenbestandes*; * = mit COUP modelliert.

Beijing area (*Mack*, 2005). The higher seepage-water amounts (185 mm y^{-1} in K *vs.* 143 mm y^{-1} in C) may lead to a corresponding increase in leaching of salt (as positive effect) and nitrate (as negative effect).

For the evaluation of the environmental impact of an irrigated plant-production system, it is essential to consider not only the soil water balance in the unsaturated zone and water-use efficiencies, but also the groundwater balance of the affected area (*Kendy* et al., 2003). Therefore, average groundwater balances for the different irrigation treatments in the field experiment and additional scenarios have been compared on a plot scale (Tab. 2, last line). Accordingly, the "optimized" sprinkler treatment B appears to be most disadvantageous because of its relatively high evapotranspiration losses. Figure 4 reveals that the groundwater balance for irrigated winter wheat production is inevitably negative under the climate conditions of the North China Plain.

3.2 Nitrate leaching—results of two recording methods

In Fig. 5, the rates of nitrate leaching under five different N-fertilization/irrigation treatments at a depth of 135 cm are presented. Since we used suction cups and passive ion-exchange samplers in parallel, we can compare the results derived from both approaches. Nitrate-leaching rates followed nearly the same pattern with respect to the relative amounts. With both methods, the combination of "conventional irrigation" and "conventional fertilization" (treatment Cc) lead to the highest amounts of nitrate leached. In contrast, the lowest nitrate leaching was observed in the treatment Ab, where suboptimal sprinkler irrigation was combined with an optimized fertilization (based on the results of N_{min} soil analysis). However, the absolute values show distinct differences. A major reason for the observed differences is the different ability of the two systems to collect percolation water moving quickly through macropores *via* preferential flows *vs.* water which slowly migrates in the soil matrix. In treatments with flood irrigation, where percolation below the rooting zone occurred at speeds of mostly >10 mm d^{-1}, nitrate amounts collected with passive samplers were clearly higher than those calculated with the suction-cup method for the same plot. This corresponds to the results of the bromide-tracer experiment, which was conducted in parallel on the same plots.

Common findings of both methods are that leached nitrate increased with higher mineral-N-fertilizer input. However, statistically (least-significant difference test) the irrigation regime can be identified as having the main effect on nitrate leaching in our experiment (*cf.*, Figure 5). Under suboptimal irrigation, nitrate leaching was very small, despite of conventional fertilization. Because of the low seepage rates under suboptimal irrigation, N accumulation in the rooting zone is most likely to occur under the conventionally fertilized treatment Ac (not shown in Fig. 5). With soil-core N_{min} analysis, *Chen* (2003) identified such nitrate-N accumulations in treatment Bc, moving downwards the 0–100 cm layer of the soil profile within one year. After flood-irrigation events and heavy rainfalls in summer, excess amounts of fertilizer N were quickly leached down below the rooting zone following preferential-flow paths. The irregular distribution of these preferential-flow paths caused a wide scatter of values between the single suction cups and passive samplers within a same plot (variation coefficients partly >50%).

Although not all N-cycling processes have been covered by measurements, the investigations by *Chen* (2003) and *Böning-Zilkens* (2004) demonstrated that an optimized N fertilization of about 70 to 100 kg ha^{-1} was sufficient to produce 5 to 6 t ha^{-1} winter wheat in our field experiment. As our measurements and calculations of nitrate leaching below the rooting zone underline, fertilization based on N_{min} soil analysis according to *Wehrmann and Scharpf* (1986) ensures a high-yield grain production in the North China Plain, while simultaneously reducing nitrate pollution of groundwater resources.

4 Conclusions

The results of our hydrological simulations suggest that an even balance between withdrawing and recharge of groundwater cannot be achieved with "optimized" irrigation by any means. In order to slow down or stop the continued decline of groundwater tables, a distinct reduction of evapotranspiration losses is necessary. This may be achieved by implying a number of plant-cultivation measures. Furthermore, available

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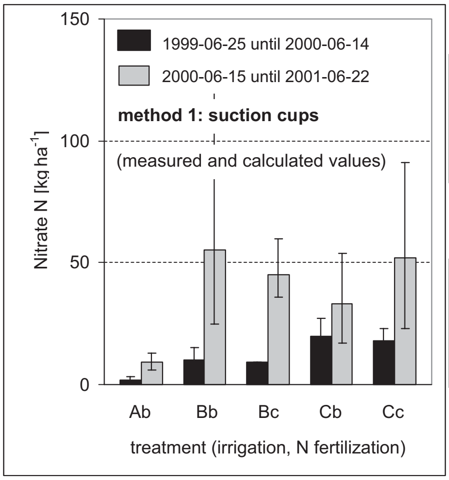


1st year:

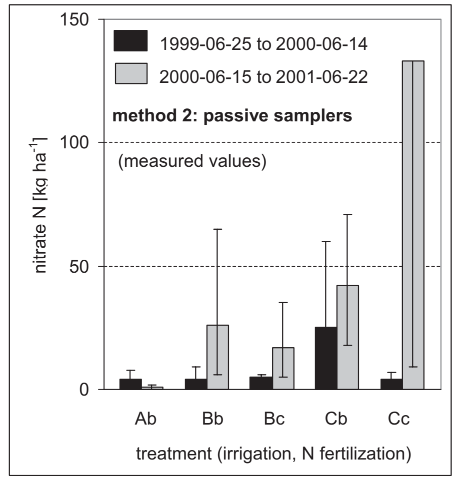
	mean	
Cb	20	a
Cc	18	ab
Bb	10	bc
Bc	9	cd
Ab	2	d

LSD (p = 0.05) = 8.2
main effect: irrigation

2nd year:

	mean	
Bb	55	a
Cc	52	a
Bc	45	ab
Cb	33	ab
Ab	9	b

LSD (p = 0.05) = 42.5
main effect: irrigation



1st year:

	mean	
Cb	25	a
Cc	5	b
Bb	4	b
Bc	4	b
Ab	4	b

LSD (p = 0.05) = 8.0
main effect: irrigation

2nd year:

	mean	
Cc	133	a
Cb	42	b
Bb	26	b
Bc	17	b
Ab	1	b

LSD (p = 0.05) = 55.9
main effect: irrigation

Figure 5: Comparison of leached amounts of nitrate N measured and calculated with two different methods at 135 cm depth; statistical significance of differences of means (LSD-test).
Explanation of abbreviations:
A: suboptimal sprinkler irrigation,
B: optimized sprinkler irrigation keeping soil moisture in the rooting zone between 45% and 80% of the available water capacity (AWC),
C: conventional flood irrigation,
a: without any N fertilization in the present cultivation period, followed by (b),
b: optimized N fertilization, based on the results of N_{min} soil analysis (70–200 kg N ha^{-1} y^{-1}),
c: conventional N fertilization in excessive amounts (600 kg N ha^{-1} y^{-1}).
Abbildung 5: Vergleich der mittels zweier Methoden bestimmten Nitrat-N-Auswaschungsmengen in 135 cm Tiefe und statistische Signifikanz der Mittelwertdifferenzen (LSD-Test).
Abkürzungserklärungen:
A: suboptimale Sprinklerbewässerung,
B: optimierte Sprinklerbewässerung zur Gewährleistung von Bodenwassergehalten zwischen 45 und 80 % der nutzbaren Feldkapazität (AWC) im Wurzelraum,
C: konventionelle Überstaubewässerung,
a: ohne N-Düngung in der aktuellen Anbauperiode, gefolgt von (b),
b: optimierte N-Düngung, bemessen anhand der Ergebnisse von N_{min}-Bestimmungen im Boden (70–200 kg N ha^{-1} a^{-1}),
c: konventionelle N-Düngung im Überschuss (600 kg N ha^{-1} a^{-1}).

water amounts can be increased by tapping water resources from areas with a surplus of water, notably in mountains.

Under the present use of N fertilizers (farmers' practice), a significant nitrate pollution of groundwater resources is taking place in the North China Plain (*Zhang* et al., 1996), notably in combination with traditional flood irrigation. As our results demonstrate, a more precise irrigation and a reduced N-fertilizer application, which follows the actual need of the crops, is feasible and at the same time ensures a high production level.

It is quite clear, however, that a model-based precise irrigation schedule requires soil physical, meteorological, and plant data, which usually are not available in practical agriculture. The same might be true for a continued soil monitoring of mineral N in the soils. Therefore, simplified but effective decision-support methods should be developed and taught to farmers. In such an effort, the Chinese Agricultural Extension Service should be involved. It is a political challenge to foster environmental awareness through education in order to achieve a careful and protective use of the soil and water resources as well as a fertilization based on crop demands. In the future, GIS applications may also be used for irrigation and fertilization management in the North China Plain (*Bareth* and *Yu*, 2002).

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